Circle drawing

Midpoint circle drawing algorithm

Initial Point (0,r) &P0=1-r

if p_k <0 the midpoint is inside the circle	if p_k >0 the midpoint is outside the circle
$x_{k+1} = x_k + 1$ • $y_{k+1} = y_k$ • $P_{k+1} = p_K + 2x_{k+1} + 1$	$x_{k+1} = x_k + 1$ • $y_{k+1} = y_k - 1$ • $P_{k+1} = p_K + 2x_{k+1} + 1 - 2y_{K+1}$

Repeat until $x \geq y$.

Ellipse drawing

Ellipse-region1

Initial Point (x0,y0)→(0,r) &P0=$r_y^2 - r_x^2 r_y + 1/4 r_x^2$

if p_k <0 the midpoint is inside the ellipse	if p_k >0 the midpoint is outside the ellipse
$x_{k+1} = x_k + 1$ • $y_{k+1} = y_k$ • $P_{k+1} = p_K + 2r_y^2 x_{k+1} + r_y^2$	$x_{k+1} = x_k + 1$ • $y_{k+1} = y_k - 1$ • $P_{k+1} = p_K + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$

Repeat until $2r_y^2 x \geq 2r_x^2 y$.

Ellipse-region2

Initial Point (x0,y0)→ Last point of region 1 &

$P0=r_y^2(x0+1/2)^2+r_x^2(y0-1)^2-r_x^2r_y^2$

if p_k <0 the midpoint is inside the ellipse	if p_k >0 the midpoint is outside the ellipse
$X_{k+1}=x_k+1$ • $Y_{k+1}=y_k-1$ • $P_{k+1}=p_K+2r_y^2x_{k+1}-2r_x^2y_{k+1}+r_x^2$	• $X_{k+1}=x_k$ • $Y_{k+1}=y_k-1$ • $P_{k+1}=p_K-2r_x^2y_{k+1}+r_x^2$

Repeat until we reach point (rx,0)

Solve the following questions

I. **Using the mid-point circle drawing algorithm, draw a circle with a radius of 6 units, along the circle octant in the 1st quadrant.**

Answer:

Initial Point (0, r) &P0=1-r

pk	x	Y
-	0	6
-5	1	6
-2	2	6
3	3	5
0	4	4

II. **Determine pixel positions along the ellipse path in the first quadrant using midpoint ellipse algorithm. Given: rx=4 and ry=3**

Answer:

- $p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$
- $p1_{k+1} = \begin{cases} p1_k + 2r_y^2 x_{k+1} + r_y^2 & p1_0 \le 0 \\ p1_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2 & p1_0 > 0 \end{cases}$
- $p2_0 = r_y^2 \left(x_0 + \frac{1}{2}\right)^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$
- $p2_{k+1} =$
  $\begin{cases} p2_{k+1} = p2_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2 & p2_0 \le 0 \\ p2_{k+1} = p2_k - 2r_x^2 y_{k+1} + r_x^2 & p2_0 > 0 \end{cases}$

Pk_region1	x_{k+1}	y_{k+1}	$2r_y^2 x_{k+1}$	$2r_x^2 y_{k+1}$
	0	3	0	96
-35	1	3	18	96
-8	2	3	36	96
37	3	2	54	64
36	4	1	72	32

Pk_region2	x_{k+1}	y_{k+1}	$2r_y^2 x_{k+1}$	$2r_x^2 y_{k+1}$
	4	1	72	32
38	4	0		